

Antiderivatives and Indefinite Integrals



Basic antiderivatives

1 Find each indefinite integral:

a $\int (3x^2 - 4x + 5) dx$

b $\int \frac{1}{x^2} dx$

c $\int \sqrt{x} dx$

d $\int \sec^2 x dx$

2 Find $\int (4\sin x + e^x) dx$.

Using initial conditions

3 Find $f(x)$ given that $f'(x) = 6x^2 - 2$ and $f(1) = 5$.

4 A particle moves along a line with velocity $v(t) = 3t^2 - 6t$ m/s. If its position at $t = 0$ is $s(0) = 4$ m, find its position function $s(t)$.

5 An object is dropped from rest, so its acceleration is $a(t) = -9.8$ m/s².

a Find the velocity function $v(t)$.

b If the object starts at height $s(0) = 45$ m, find its position function $s(t)$.

c Find how long it takes to hit the ground ($s = 0$), to two decimal places.

More antiderivatives

6 Find each indefinite integral:

a $\int (\frac{1}{x} - 3x^{-4}) dx$

b $\int (2\sec x \tan x) dx$

c $\int \frac{4}{\sqrt{1-x^2}} dx$

7 Find the general antiderivative of $f(x) = (x + 1)^3$ (expand first, or use the power rule on the whole expression, and state which method you used).

Motion applications

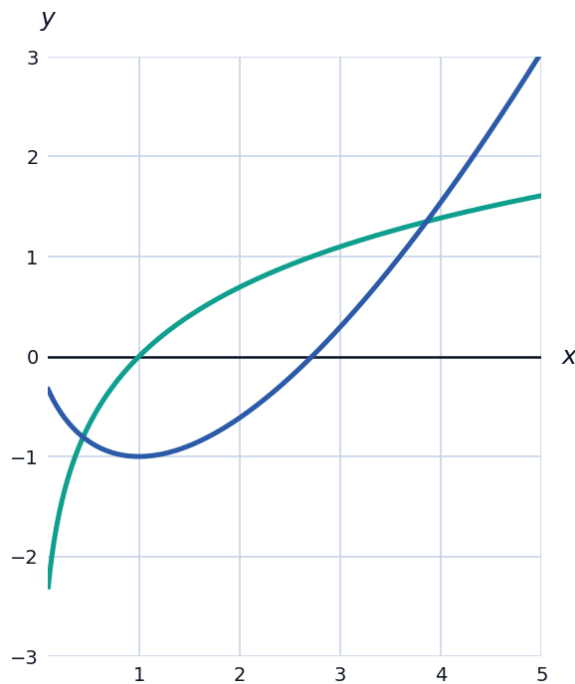
- 8 A rocket's acceleration is $a(t) = 6t \text{ m/s}^2$, with $v(0) = 0$ and $s(0) = 0$.
- a Find $v(t)$.
 - b Find $s(t)$.
 - c Find the distance traveled in the first 4 seconds.
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Antiderivatives of composite expressions

- 9 Find $\int (2x - 1)^5 dx$, treating $(2x - 1)$ as a linear inner function.
- 10 Find $\int \sin(3x) dx$ and $\int e^{5x} dx$.
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Verifying antiderivatives by differentiation

- 11 Verify that $F(x) = x \ln x - x$ is an antiderivative of $f(x) = \ln x$ by differentiating $F(x)$.



Antiderivatives of reciprocal and rational-power functions

12 Find each indefinite integral:

a $\int \frac{3}{x} dx$

b $\int x^{-2/3} dx$

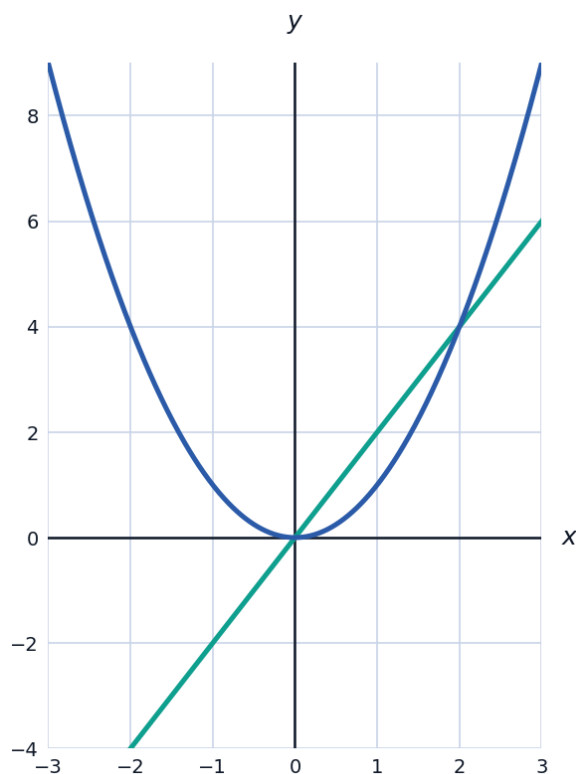
c $\int \frac{1}{2\sqrt{x}} dx$

A projectile motion application

13 A ball is thrown upward with initial velocity 15 m/s from a height of 2 m, so its acceleration is $a(t) = -9.8$ m/s². Find its velocity function $v(t)$ and position function $s(t)$.

Visualizing an antiderivative family

14 The graph shows $f(x) = 2x$ and one member of its antiderivative family, $F(x) = x^2$. Every other antiderivative of f is a vertical shift of F .



a Write the general antiderivative of $f(x) = 2x$.

b If a specific antiderivative passes through $(1, 4)$, find its constant C .

Antiderivatives requiring algebraic rewriting first

15 Find $\int \frac{x^3 + 2x}{x} dx$ by simplifying the integrand before integrating.

16 Find $\int (x + 1)(x - 3) dx$ by expanding the product first.